

Explainable AI for Academic Integrity Verification

Progress since Meeting 4, and the run-in to the graded presentation

Mykhailo Shpyl | MSc AI and Big Data Analytics, ATU
Supervisor: Dr. Vini Vijayan | Meeting 5, 7 July 2026

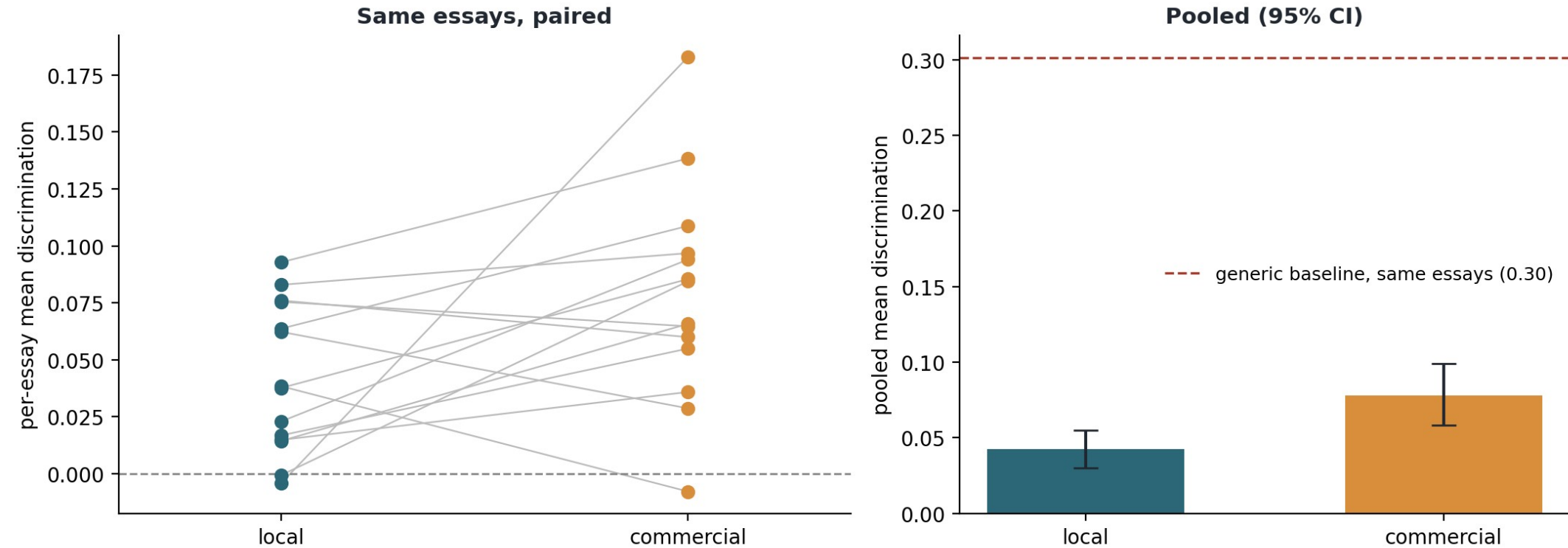
Since Meeting 4 (23 June): the whole pipeline now runs

- Every locked-scope component is trained: the hybrid detector, the explainability layer, the claim extractor, the Bloom classifier, both question-generation backends, and the output assembler.
- The commercial-vs-local comparison ran twice: at scale, then on a tighter fixed-claim design that changed the answer.
- The fine-tune ran on your sign-off, its first result was a trap I caught by reading the output, and a controlled three-step experiment fixed it.
- The three-judge LLM evaluation is complete, and the detector is now the full hybrid (component 1 finished this week).
- The pipeline assembles the lecturer's guide end to end from every trained component.
- Writing: 78-page draft, now including the Discussion and Conclusions chapters; 33 references, each verified against the real paper.

Commercial vs local: the first answer

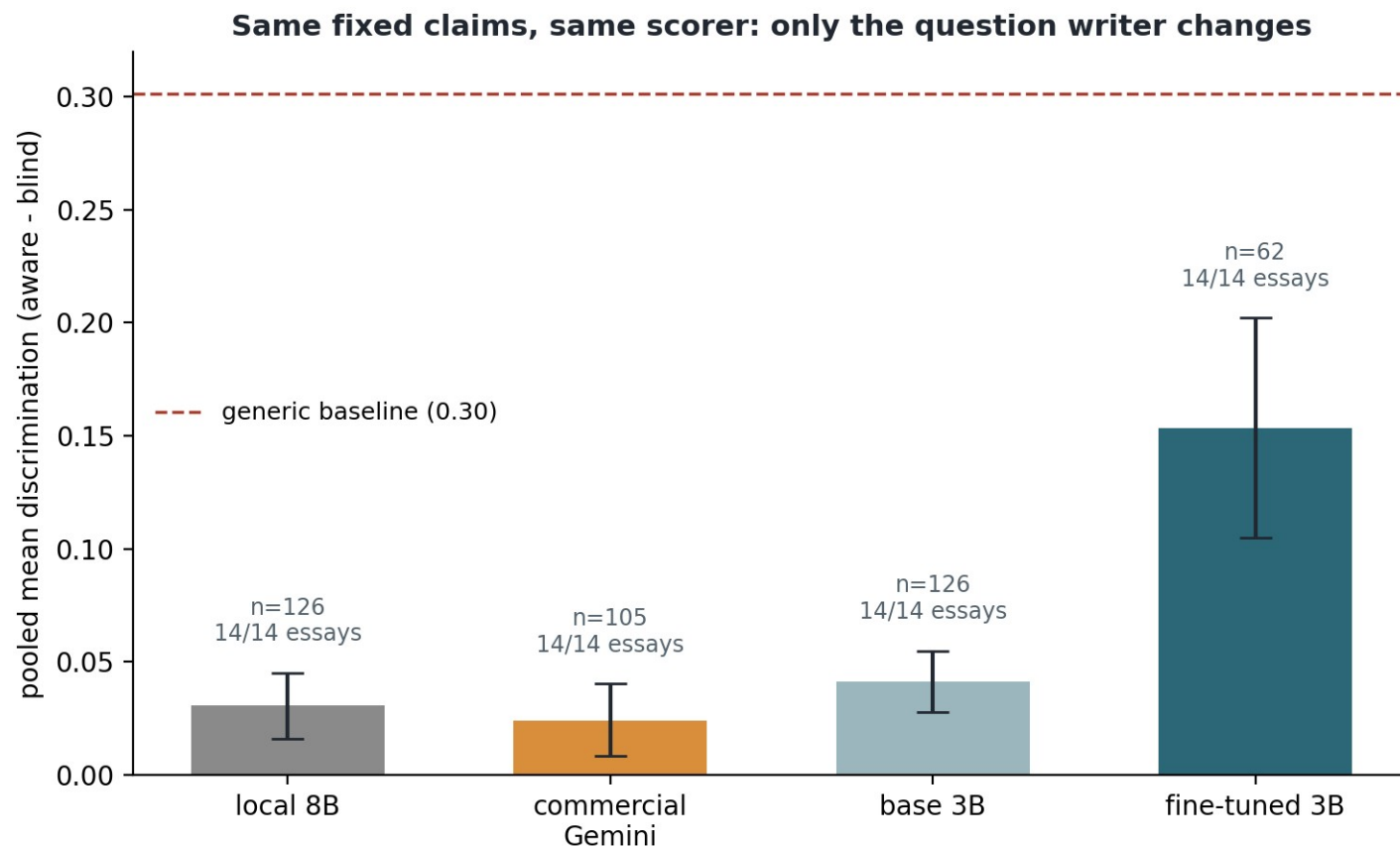
Fourteen essays, 126 questions per backend, identical scoring. Local Llama 3.1 8B 0.042 [0.030, 0.055] vs free-tier Gemini 0.078 [0.058, 0.098]. Paired difference 0.036 (t p = 0.040, Wilcoxon p = 0.030): a small but significant commercial edge, from a local model not yet fine-tuned.

Question generation: commercial vs local across 14 essays



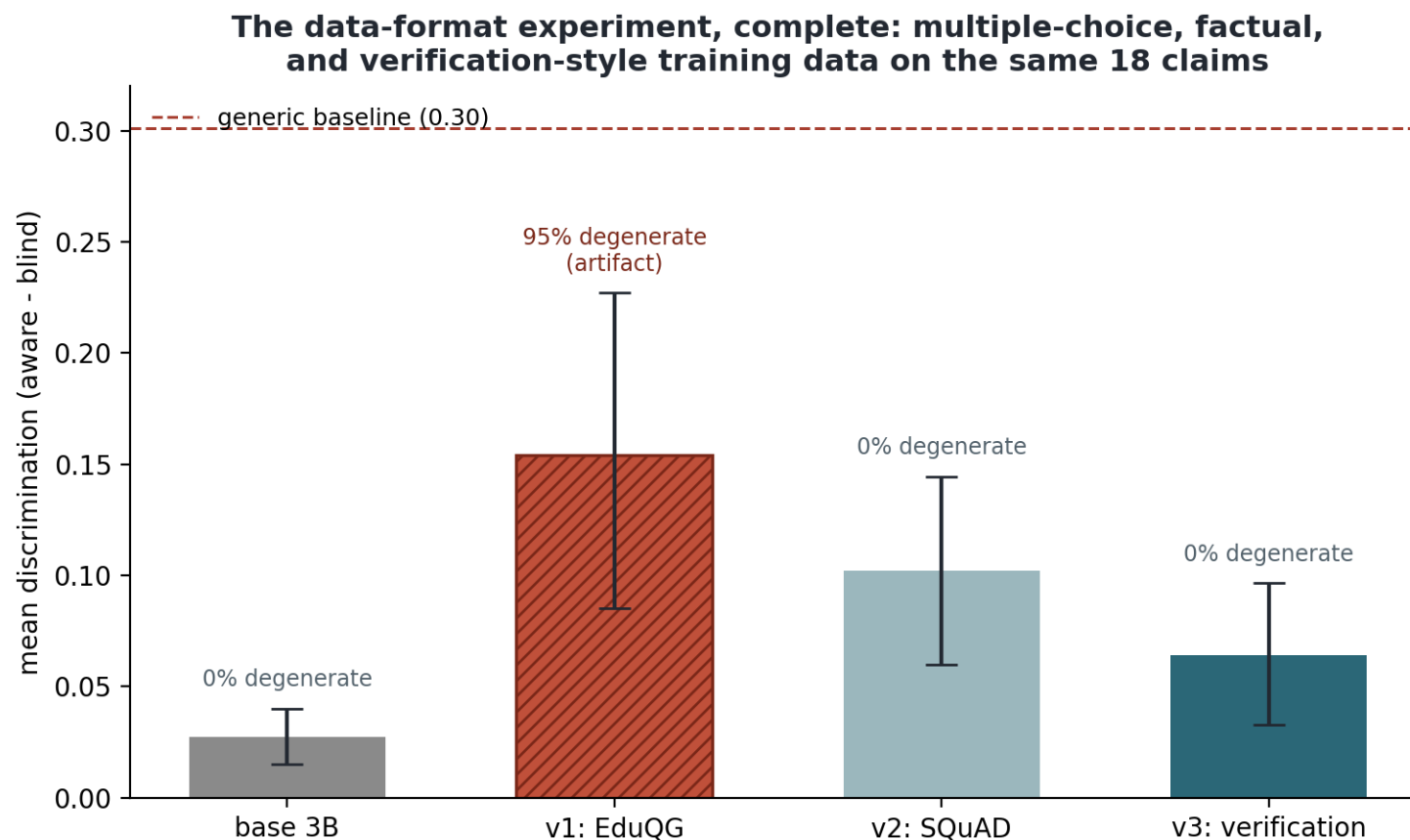
The tighter test: the commercial edge disappears

One fixed claim set per essay (neutral extractor), every writer answering the SAME claims, 14/14 essays. The edge from the previous slide vanishes: Gemini 0.024 vs Llama 8B 0.031 and base 3B 0.041, paired difference -0.005, $p = 0.62$, higher on 7 of 14, a coin flip. It came substantially from claim selection, not from writing better questions. The tall bar is the fine-tuned model, and the next slide is why it cannot be read at face value.



The result that lied, and the experiment that fixed it

The fine-tune, controlled across three training sets, everything else identical. v1 on EduQG scored 0.154 (tall) but 95% of its questions were degenerate multiple-choice stems that game the metric. v2 on SQuAD: 0% degenerate, a real gain (0.102). v3 on verification-style questions self-distilled from the pipeline: 0% degenerate, on-style, 0.064, double the 8B teacher. The metric alone would rank the unfit v2 first, so it does not decide alone.



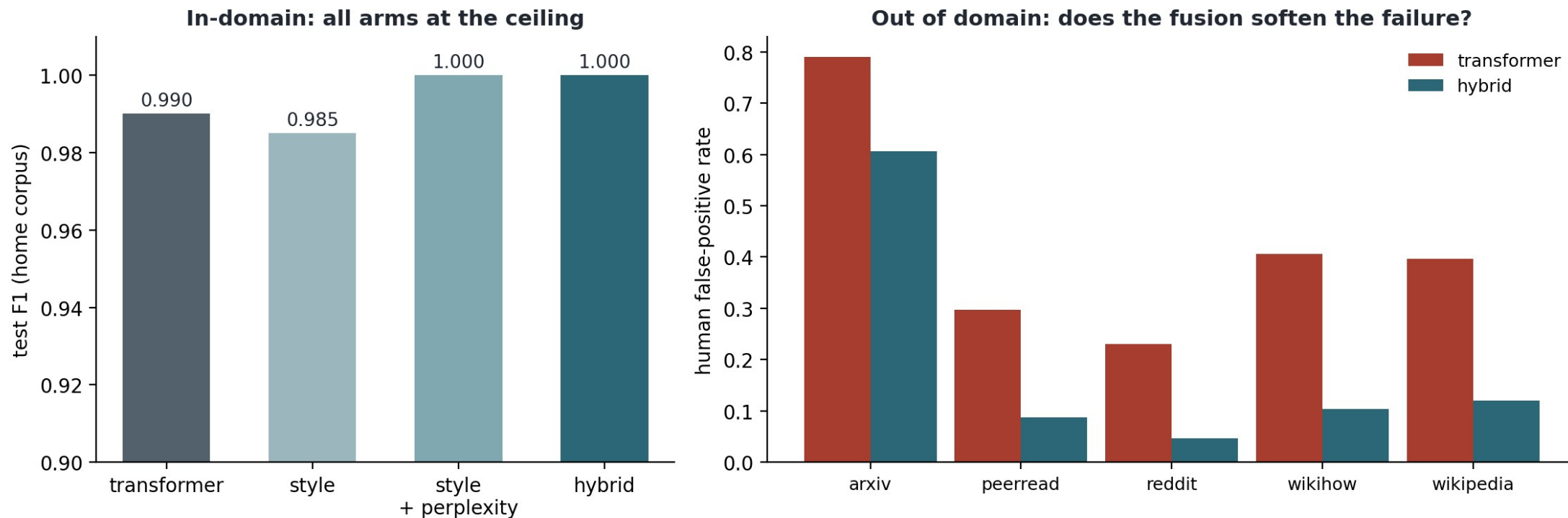
Three judges say the same thing: do not trust one number

- The scope's supplementary check: three commercial LLMs (Gemini, Claude, GPT) rate each question on a four-part rubric, 1 to 5.
- Two sit at the ceiling (means 4.81 and 4.94); the third uses the scale (3.67). They do not agree: Krippendorff's alpha -0.25, no pairwise correlation significant.
- None tracks the objective simulation; GPT's near-ceiling ratings are even significantly negative (rho -0.75).
- Reading: rubric scores measure how good a question looks, not whether it works. This is the second independent time an automatic score pointed the wrong way, after the fine-tune.
- So the judge-free simulation carries the evidence, exactly as the evaluation plan intended, and every judge is anchored to it.

The detector, finished as a hybrid, and made fairer

Component 1 complete: the transformer fused with the stylometric features and GPT-2 perplexity. In-domain everything is at the ceiling (perplexity is the single strongest feature). Out of domain the fusion barely moves F1 but cuts the human false-positive rate in every domain, by three to eight times in four of five (arXiv 0.79 to 0.61, reddit 0.23 to 0.05): fewer false accusations, which is the failure that matters.

The hybrid detector (component 1 complete): transformer + style + GPT-2 perplexity



The finished artifact: every trained component in one guide

The Verification Interview Guide, generated end to end with live models on the laptop, in the design you approved. Hybrid detector verdict with its component views; each claim in readable phrasing anchored both to its cited sentences and to the trained miner's verbatim argument spans; questions from the fine-tuned backend, gated for quality and tagged by the trained Bloom classifier; opening line: evidence for a conversation, not an accusation.

Verification Interview Guide

Submission: 3108a | **Generated:** 2026-07-07 | **Question backend:** local-hf:v3:qg_finetune_qwen3b_v3

This guide is evidence for a conversation with the student, not an accusation. The detector's judgement is fallible, and the fair use of this document is to ask, listen, and weigh the answers.

1. Detection summary

The hybrid detector (transformer fused with stylistic features and GPT-2 perplexity) scores this submission at **0.9572** probability of being AI-generated, so it is **flagged as likely AI-generated**. No score of 1.0 exists on this scale: the model is never certain, only confident. On matched in-domain test data the detector's F1 is 0.99; on out-of-domain academic text its false-positive rate rises sharply, so treat the score as a reason to talk, never as proof.

Component views: transformer 0.9996, style-plus-perplexity 0.9985. The hybrid fuses both because the style half keeps the detector calmer on unusual but human writing.

What drives decisions like this one (validated by faithfulness testing):

- Average word length: consistently longer words push a text toward the AI class.
- Vocabulary richness: a wider, less repetitive vocabulary pushes toward the human class.
- Sentence-length variation: humans vary sentence length more; uniform sentences look AI-written.
- Auxiliary-verb and determiner densities: small grammatical habits that differ between the classes.

2. The student's claims and where they come from

Each claim below is phrased for readability, then anchored two ways so nothing here is invented: to the exact sentence numbers it was drawn from, and to the verbatim argument spans the trained argument miner found in those sentences, each labelled by its role (major claim, claim, or premise). Every question can be traced to the student's own words.

Claim 1: The three texts (Dickinson's poetry, Shakespeare's Othello, and Austen's Emma) represent distinct literary genres but share certain commonalities in their exploration of universal themes and ideas.

Source in the submission (cited sentences):

- [1] These three texts represent distinct literary genres: poetry, drama, and prose fiction respectively. [27] Despite these differences, all three texts share certain commonalities. [31] These texts represent distinct literary genres but share certain commonalities in their exploration of universal themes and ideas.

Writing: closed out and grown to a full draft

- Your Meeting 4 feedback stays closed: overview-first introduction, elaborated background, per-chapter outline, bullet-spacing bug fixed in the builder.
- The draft is 78 pages, eleven chapters. New this fortnight: the Discussion (9) and Conclusions (10) chapters, the hybrid detector (6.7), the training-distribution control (6.6), attention as the third explanation method (5.6), and the full fine-tune and judge story (8.7 to 8.11).
- 33 references, each verified against the real paper before it went in (arXiv, ACL Anthology, NeurIPS, CrossRef, publisher PDFs); in-text citations placed at first use.
- A staleness pass so no chapter promises work that is already done.
- You have the Word version; every number in it traces to a saved result file.

What is left, and one decision for you

- Integrate and demo done: the approved spans-plus-phrasing guide runs end to end (previous slide).
- Small and mine: the promised detector retrain to strip one residual token; appendices (the full guide, the exact prompts, a hyperparameter table); a final numbers-and-citations audit.
- Your call: the scope mentions relation classification between claims and premises alongside the span tagger. The question generator only needs the claims, so I propose either a basic version (about two days) or a recorded descope. I am comfortable either way.
- Plan: full draft to you by end of July for a feedback round; August protected for writing, revision and submission.
- Today, most usefully: a ten-minute run-through of the graded 14 July presentation.